

Cryptography - 23_09

The exam is written, with no notes or books. It is divided in three parts, with each part being one exercise and one theory questions.

The recommended book is "Katz Lindell, Introduction to Modern Crypto".

🔍 Question

What is this course about?

It's an introduction to modern cryptography.

Why modern? Because the way crypto is done today is different from the past.

It exists for millennia. This course will focus on secure communication.

☰ Example

One famous example is the Ceaser Cypher. With it, letters are transposed of a certain offset, like $A \rightarrow C$, $B \rightarrow D$, with offset = 2.

This example is **not** a good cryptcho algorithm, it's easy to break.

Before the '50: Crypto was an art. It was secure until not broken.

☰ Famous Example: TLS

Modern crypto, in the other hand, means Crypto as a science.

- Precise definitions.
- Proofs: there would be two "flavors":
 - Unconditional proofs: no conditions (assumptions). Possible, but inefficient.
 - Conditional proofs: there are assumptions. Like $P \neq NP$.

There exists problems that seem to be hard for efficient computation.

An example is **Factoring**, where $n = p \cdot q$.

We have p, q as prime numbers, $|p| \approx |q| \times \lambda$ *bits*, and λ as a secret parameter.

It is hard to compute p, q given n , that's an assumption.

A recipe for provable security. Prove this:

✓ Theorem

Cryptosystem X is "secure" if factoring is hard.

ⓘ Proof

Assume X not secure: $\exists$ efficient machine "breaking" X .

Then $\exists$ efficient machine $\mathbb{B}$ solving factoring.

■ (was an example)

Secure Communication

We have Alice and Bob, the first goal of the course is that Alice wants to send a message $m \in M$ to Bob.

Whatever message is sent, it can also be seen by a third party, Eve.



With the second goal we don't care about the secrets of the message. We talk about message integrity, we don't want the message to be modified by a third party.



We will cover two approaches:

- Symmetric Crypto: Alice and Bob share a common key $k \in K$ (K is the key space), the key is random and unknown to Eve.
- Asymmetric Crypto: Alice and Bob do not share a common key, but they have each their own key pair (p_k, s_k) where p_k is public and s_k is secret.

Unconditional Security

We start with goal 1 in symmetric crypto.

We have Alice and Bob sharing a key $k \in K$, Eve doesn't know it.

Alice sends a message to Bob using a box $\boxed{Enc}$ (with the key k) and its output is the content sent, $c \in C$, where c is the cyphertext. Bob has a decryption box $\boxed{Dec}$, where he uses the key to decrypt c and get m .

✓ Definition

Symmetric Encryption SKE : $\Pi = (Enc, Dec)$ such that:

- $Enc : M \times K \rightarrow C$
- $Dec : C \times K \rightarrow M$
- k is uniform over K .

🔗 Correctness

$$\forall k \in K, \forall m \in M \quad Dec(K, Enc(k, m)) = m$$

If I decrypt with key k the cyphertext $c = Enc(k, m)$, i'll get m back.

🔗 Kerchoff

"Security only should depend on secrecy of the key, not on algorithm".

We then have Shannon (1950): Put forward a definition called **perfect secrecy**.

✓ Definition

Let M be any distribution over $\mathcal{M}$ and K be uniform over $\mathcal{K}$. (Then observe that $C = Enc(K, M)$ is a distribution over $\mathcal{C}$).

We say that $(Enc, Dec) = \Pi$ is perfectly secret if $\forall M, \forall m \in \mathcal{M}, \forall c \in \mathcal{C}$
 $Pr[M = m] = Pr[M = m | C = c]$.

(The probability that the message M is m is equal to the probability that

knowing that $M = m$ knowing that $C = c$, so knowing the ciphertext doesn't do anything at all.)

🔗 Intuition

Ciphertext C reveals nothing on plaintext m , besides what you knew already.

Shannon did two things:

1. The definition is achievable
2. It comes with inherent limitations in practice.

🔗 Lemma

The following are equivalent:

1. Perfect secrecy.
2. M and C are independent.
3. $\forall m, m' \in M, \forall c \in C, \Pr_K[Enc(K, m) = c] = \Pr_K[Enc(K, m') = c]$

With $K \equiv$ uniform over $\mathcal{K}$, and $\Pr_K$ is the probability over K (so uniform).

Now we'll see how to get perfect secrecy.

The one-time pad: $K = \mathcal{M} = \mathcal{C} = \{0, 1\}^n$

$$Enc(k, m) = k \oplus m$$

$$Dec(k, c) = c \oplus k = (m \oplus k) \oplus k = m$$

✓ Theorem

OTP is perfectly secret.

📖 Proof

We use definition three in the above lemma.

Well, for any $m, m' \in M$ and $c \in \mathcal{C}$:

$$\Pr[Enc(K, m) = c] = \Pr[k \oplus m = c] = \Pr[k = m \oplus c] = 2^{-n}.$$

For the same reason

$$\Pr[Enc(k, m') = c] = 2^{-n}$$



There seem to be limitations:

- The key is as long as the message.
- Key can only be used once.

In fact, assume we encrypt m_1, m_2 with k . Then:

$$c_1 = k \oplus m_1; c_2 = k \oplus m_2$$

Then $c_1 \oplus c_2 = m_1 \oplus m_2$, *the key is gone*.

If I know single pair (m_1, c_1) we can compute m_2 .

For Shannon, you cannot encrypt two messages with the same key, even though we obviously want to do it.

✓ Theorem (Shannon)

Let Π be any perfectly secret SKE, then

$$|\mathcal{K}| \geq |\mathcal{M}|$$

📖 Proof

Take M to be uniform over $\mathcal{M}$. Take also any c such that $\Pr[C = c] > 0$ (*it appears sometimes*), with $C = Enc(k, M)$.

Consider $\mathcal{M}' = \{Dec(k, c) : k \in \mathcal{K}\}$ (*all possible ways we can decrypt c*)

Assume $|\mathcal{K}| < |\mathcal{M}|$ by contradiction.

Then:

$$|\mathcal{M}'| \leq |\mathcal{K}| < |\mathcal{M}| \text{ (it's an upper bound)}$$

$$\implies |\mathcal{M}'| < |\mathcal{M}|.$$

Then there must be something that's in $\mathcal{M}$ but not in $\mathcal{M}'$ ($\exists m \in \mathcal{M}/\mathcal{M}'$).

Now:

$$\Pr[M = m] = \frac{1}{|\mathcal{M}|} \text{ (it's uniform)}$$

On the other hand:

$\Pr[M = m | C = c] = 0$ (because $\mathcal{M}'$ contains all possible decryptions of c , so it can't be outside $\mathcal{M}'$).

This proves that they are not (?) different.

